Assignment 1 - Discrete Mathematics (Monsoon 2025)

"Each problem that I solved became a rule which served afterwards to solve other problems."
- by René Descartes

Submission deadline: 11:59 PM on 9th September, 2025. **Maximum marks**: 60

Instructions. Consider the following while preparing the answers for the given questions.

1. Kindly go over the academic dishonesty policy of IIIT Delhi given on this link. Any deviation from this policy will lead to the strictest possible punishment.
2. Each question is of 10 marks.
3. It is prohibited to use AI-related tools like chatGPT to solve the questions, unless explicitly mentioned. Any violation in this regard will be handled very strictly.
4. You are encouraged to write solutions in LaTeX. Anyone who submits homework in LaTeX(i.e., in the pdf format and not the source code) will get an additional 20% of their score in that homework.
5. While writing the solutions, you can refer to the topics, theorems, examples, etc., taught in the class. It is not allowed to refer to anything related to propositional and predictive logic not covered in the class.
6. Late submissions are not allowed.
7. If you discuss anything related to the solution of a question in this assignment with anyone, you must mention their names.

1 Problem Set

In this document, $\equiv$ means equivalence of propositions.

Question 1.1. *1. (6 marks). Let p, q, r be propositional variables. Write the truth tables of the following propositions.*

(a) $\neg(p \oplus q) \implies \neg q \vee r$.

(b) $p \vee \neg r \iff q \wedge r$.

(c) $(p \oplus \neg q) \implies q \wedge \neg r$.

(d) $p \wedge (p \iff q) \implies \neg q$.

2. (2 marks). Convert the following to predicates with quantifiers and then write the negations of these quantified predicates.

(a) There exists a student at IIIT Delhi who knows swimming and driving.

(b) Every fruit in the basket is either a mango or a strawberry.

3. (2 marks). Let x be a variable and $P(x)$ be a predicate. Explain why $\forall x P(x)$ and $\exists x P(x)$ are propositions?

Question 1.2. *Let p, q, r be propositions. State whether the following are true, false, or neither. Justify your answers.*

1. $p \oplus \neg p \equiv p$.
2. $\neg(p \vee q \vee r) \equiv \neg p \wedge \neg r \vee q$.
3. $p \iff p \equiv \neg p \oplus \neg p$.
4. $p \implies (p \wedge (p \vee p)) \equiv (p \vee \neg p)$.

5. $(p \iff \neg p) \equiv p$.

6. Suppose there exists a student in the CS department who owns a computer, there are at least two students in the CSE department and Ram is one of them. Then, Ram owns a computer.

7. Let x be a variable and $f = x^2 + 4x - 5$ be a polynomial. Then, there exists a natural number a such that $f(a) = 0$.

8. Let x be a variable and $P(x), Q(x)$ be predicates. Then, $\neg\exists x(P(x) \implies \neg Q(x)) \equiv \forall x(P(x) \wedge Q(x))$.

9. Suppose every student in IIIT Delhi has a phone. Then, Annirudh also has a phone.

10. Let x be a variable and $P(x), Q(x)$ be predicates. Then, $\neg\forall x(P(x) \wedge \neg Q(x)) \equiv \exists x\,(\neg P(x) \vee Q(x))$.

Question 1.3. *Let p, q, r be propositional variables. Do the following by using only the algebra of propositional logic. You must mention the rule of algebra at every step.*

1. *Convert $(\neg(p \oplus \neg q) \iff r)$ to an equivalent CNF.*
2. *Convert $\neg(p \wedge (q \oplus r))$ to an equivalent DNF.*
3. *Convert $(p \implies \neg q) \oplus (q \iff (r \implies \neg r))$ to an equivalent conjunctive form (need not be a CNF).*
4. *Convert $(p \iff \neg q) \implies (q \oplus (p \wedge \neg\neg r))$ to an equivalent disjunctive form (need not be a DNF).*

Question 1.4. *Let u be an arbitrary propositional formula defined using propositional variables $p_1, \ldots, p_n$.*

1. *(3 marks). Let v be the DNF obtained from the truth table of u by following the procedure taught in the class. Prove that $u \equiv v$.*
2. *(6 marks). Let y be the CNF obtained from the truth table of u by following the procedure taught in the class. Prove that $u \equiv y$.*
3. *(1 mark). Why don't we have to prove the above-mentioned equivalence if v, y are constructed by following the procedures for constructing CNFs and DNFs using the algebra of propositional logic?*

Question 1.5. 1. *(5 marks) Let $p_1, p_2, p_3, p_4, p_5, p_6$ be propositional variables. Let u be the propositional formula defined as*

$$u := (\neg p_1 \vee p_2) \wedge (\neg p_3 \vee p_4) \wedge (\neg p_2 \vee p_3) \wedge (\neg p_5 \vee p_6) \wedge (\neg p_6 \vee p_1) \wedge (\neg p_4 \vee p_5).$$

How many $a_1, a_2, a_3, a_4, a_5, a_6$ exist such that every a_i is either true or false, and when we substitute p_i with a_i in u for every $1 \le i \le 6$, we get true? Justify your answer.

(Hint: Can commutativity help to nicely restructure u?)

2. *(5 marks). Let u be an arbitrary CNF and v be an arbitrary DNF in propositional variables $p_1, \ldots, p_n$.*
   (a) *Does there exist an OR-term w_1 containing variable p_i for every $1 \le i \le n$, with or without negation such that w_1 is not present in u and $u \wedge w_1 \equiv u$? Justify your answer.*
   (b) *Does there exist an AND-term w_2 containing variable p_i for every $1 \le i \le n$ with or without negation such that w_2 is not present in v and $v \vee w_2 \equiv v$?*

We state some definitions for the last question.

Definition 1 (Covering portion of an AND-term)**.** *Let $p_1, \ldots, p_n$ be propositional variables and u be an AND-term in some variables $p_1, \ldots, p_k$ for some natural number $k < n$. Let v be the proposition*

$$v := (u \wedge \neg p_{k+1} \wedge \cdots \wedge \neg p_{n-1} \wedge \neg p_n) \vee (u \wedge \neg p_{k+1} \wedge \cdots \wedge \neg p_{n-1} \wedge p_n) \vee \cdots \vee (u \wedge p_{k+1} \wedge \cdots \wedge p_{n-1} \wedge p_n),$$

i.e., v is the OR of the AND-terms constructed using u by introducing every remaining variable $p_{k+1}, \ldots, p_n$ not appearing in u, once with negation and other time without negation. In other words, v is the DNF equivalent to the AND-term u. Then, v is said to be the covering portion of u.

For example, $(p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$ is the covering portion of the AND-term $(p \wedge q)$.

Definition 2 (Cover of a disjunctive form)**.** *Let u be a disjunctive form and v be a DNF in propositional variables $p_1, \ldots, p_n$. Then, v is said to be the cover of u if every AND-term of u has a covering portion in v.*

For example, let $u = (p \wedge q) \vee (p \wedge r)$ and $v = (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r)$. Then, it is easy to verify that v is a cover of u.

Similarly, we can define the covering portion of an OR-term and the cover of a conjunctive form.

Question 1.6 (Uniqueness of DNFs).

1. *(2.5 marks). Let u_1 be an AND-term in the propositional variables $p_1, \ldots, p_n$. Let v_1 be the covering portion of u_1. Prove that $u_1 \equiv v_1$.*

   (Hint: Use the rules of algebra of propositional logic).

2. *(2.5 marks). Let u_2 be a disjunctive form in propositional variables $p_1, \ldots, p_n$. Let v_2 be the cover of u_2. Prove that $u_2 \equiv v_2$.*

   (Hint: take help from the first part of this question and use the definition of a disjunctive form).

3. *(2.5 marks). Let u_3, u_4 be disjunctive forms in the propositional variable $p_1, \ldots, p_n$. Let v_3, v_4 be the covers of u_3, u_4, respectively. Suppose $u_3 \equiv u_4$. Prove that v_3 and v_4 are the same DNFs.*

   (Hint: Take help from the second part of this question and the second part of Question 1.5).

4. *(2.5 marks). Let w be an arbitrary proposition in variables $p_1, \ldots, p_n$. Prove without using the truth table of w that there exists a unique DNF that is equivalent to w.*

   (Hint: What will you do if you get two different equivalent disjunctive forms from the third-last step of the procedure to construct DNF using the algebra of propositional logic? Does any part of this question help?)

Remark. Similar to Question 1.6, we can also show using the same ideas that every proposition has a *unique* equivalent CNF.